UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Important points

Definitions, statements, key points, "why" reasoning and exam traps, topic by topic.

Class 001 · Introduction to partial differentiation · compiled 15 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S001·15	Deck 001, slide 15 (rendered .pptx)
T001a 00:14:20	Transcript "Introduction to Partial Differentiation", segment start
T001b 00:11:41	Transcript "Continuity and differentiability of a function of two independent variables"
CW U1 P7	Unit 1 class-work problem 7
ESA Dec-23 1a	Past paper: Dec 2023, question 1.a

Contents

1. 1 · Functions of two or more variables
2. 2 · Limit of a function of two variables
3. 3 · Continuity of a function of two variables
4. 4 · Partial derivatives: history, definition and notation
5. 5 · Second-order and mixed partial derivatives
6. Past-paper questions

1 Functions of two or more variables

Definitions

Function of two variables. A function of two variables denoted by $z = f(x, y)$ is a rule that assigns to each ordered pair of real numbers (x, y) in a set D a unique real number denoted by $f(x, y)$. The set D is the domain of f and its range is the set of values that f takes on: $\{f(x, y) : (x, y) \in D\}$.

S001·8 · T001a 00:05:46

Independent and dependent variables. In $z = f(x, y)$, x and y are independent variables and z is the dependent variable (a bivariate function); with three independent variables one writes $w = f(x, y, z)$, and in general $f(x_1, x_2, \ldots, x_n)$. TRANSCRIPT ONLY T001a 00:02:18 · T001a 00:02:54

Key points

1. Examples on the slides: area of a rectangle $A(l, b)$; volume of a cylinder $v(r, h)$; electric potential in a cathode-ray tube $V = f(x, y, z, t)$; propagation of an electromagnetic wave (magnetic field, electric field, wavelength, direction); weather (topography, pressure, humidity, temperature); spread of a pandemic (area, population, awareness, immunity, age and gender distribution). ESA ★ S001·4 · S001·5 · S001·6 · S001·7 · T001a 00:03:26
2. A single-variable function is enough to understand a concept, but to draw a conclusion about a real situation we almost always need several variables. ESA ★ (transcript only) T001a 00:05:16
3. f maps an ordered pair to one real number: for $z = x^3y$, the pair $(2, 1)$ is mapped to 8. ESA ★ T001a 00:07:16
4. Graph of $z = f(x, y)$: a surface, which needs three axes; a function of three variables would need four dimensions, so the unit sticks to two variables. The slide shows $\sin x \cos y$ and $e^{1-x^2-y^2}$ (Figures G1.1–G1.2). ESA ★ S001·8 · T001a 00:08:35 · T001a 00:09:07

2 Limit of a function of two variables

Definitions

Limit of a function of two variables. A function $f(x, y)$ is said to approach the limit L as (x, y) approaches (a, b), if the limit remains the same along any path of (x, y) approaching (a, b): $\lim_{(x,y)\to(a,b)} f(x, y) = L$. S001·9 · T001a 00:11:01

Key points

1. One variable: $x \to a$ from the left or the right only. Two variables: $(x, y) \to (a, b)$ along any of infinitely many paths (Figure G2.1). ESA ★ S001·9 · T001a 00:11:32 · T001a 00:12:34
2. If substitution gives a number (a polynomial, no $\frac{0}{0}$), that number is the limit, whatever the path. ESA ★ T001a 00:13:59
3. It is hard to show that a limit exists but easy to show it does not: two paths with different values are enough. ESA ★★ S001·10 · T001a 00:17:42
4. Agreement along a few paths proves nothing: for $\frac{2xy}{3x^2+y^2}$, the x-axis and the parabola $y = x^2$ both give 0, but $y = x$ gives $\frac{1}{2}$. ESA ★★ T001a 00:16:48 · T001a 00:17:24

Why?

Why can one path never prove that a limit exists?
There are infinitely many paths, and a function can agree on any finite number of them and still differ on another: $\frac{2xy}{3x^2+y^2}$ gives 0 on the x-axis and on $y = x^2$ but $\frac{1}{2}$ on $y = x$. To prove existence one needs an estimate for all paths at once (e.g. polar coordinates: $\frac{3x^2y}{x^2+y^2} = 3r\cos^2\theta\sin\theta \to 0$).
ADDED FOR CLARITY T001a 00:17:42 · S001·12

Exam traps

Exam trap. Same value along two or three paths does not prove that the limit exists; different values along two paths does prove it does not. T001a 00:16:48 · T001a 00:17:42

Exam trap. Slide 001·9 uses the point $(2, 1)$ (limit -8); the lecture uses $(2, -1)$ (limit -6). Read the point in the question. ADDED FOR CLARITY S001·9 · T001a 00:12:59

3 Continuity of a function of two variables

Definitions

Continuity at a point. $f(x,y)$ is continuous at (a,b) if $\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$: $f(a,b)$ must be defined, the limit must exist, and the two must be equal. S001·11 · T001b 00:00:38 · T001a 00:18:46

Statements in the class material

> *"A function is continuous at a point if the graph doesn't have any holes or breaks at that point. Graphing the function is helpful. If a function is not continuous at (a,b), this does not mean the limit does not exist. We will need to consider the approaching (a,b) from different paths."* S001·11

Key points

1. One variable: f is continuous at c if (i) $\lim_{x\to c} f(x)$ exists, (ii) $f(c)$ exists, (iii) they are equal. The same three conditions define continuity for $f(x,y)$. ESA ★ T001a 00:18:46 · S001·11
2. $\frac{3x^2y}{x^2+y^2}$: the limit at the origin exists (it is 0) but $f(0,0)$ is $\frac{0}{0}$, so f is not continuous there. $\frac{2xy}{3x^2+y^2}$: the limit does not exist and $f(0,0)$ is undefined (Figures G3.1–G3.2). ESA ★ S001·12 · S001·13 · T001b 00:04:14 · T001b 00:05:44
3. To decide continuity in higher dimensions, graph the function: a hole, fold or sudden drop in the surface shows a discontinuity. ESA ★ S001·11 · T001b 00:06:45 · T001a 00:18:07

Why?

Why is $\frac{3x^2y}{x^2+y^2}$ not continuous at the origin even though its limit is 0?
Continuity needs $f(0,0)$ to be defined and equal to the limit. Here $f(0,0)$ is $\frac{0}{0}$, so the third condition cannot be checked. Defining $f(0,0)=0$ would make it continuous.

ADDED FOR CLARITY S001·12 · T001b 00:04:14

Exam traps

Exam trap. "Not continuous" does not mean "the limit does not exist" (slide 001·11): $\frac{3x^2y}{x^2+y^2}$ has limit 0 at the origin. S001·11 · T001b 00:02:10

Exam trap. Continuous is not the same as smooth: $z=|x|+|y|$ is continuous everywhere but has a corner at the origin. The lecture's "continuous if the surface is smooth" is loose.

ADDED FOR CLARITY T001b 00:06:26

4 Partial derivatives: history, definition and notation

Definitions

Partial derivative. A partial derivative of a function of several variables is the derivative with respect to one of those variables, with the others held constant. S001·15 · T001b 00:12:52

Key points

1. The product rule is already partial differentiation: in $\frac{d}{dx}(x^3 \sin x) = x^3 \cos x + 3x^2 \sin x$, each term treats the other factor as a constant k. Leibniz: $d(uv) = u\,dv + v\,du$. ESA ★ S001·14 · T001b 00:07:56 · T001b 00:10:18
2. History: Leibniz first used partial differentiation (product rule); Adrien-Marie Legendre created the modern symbol ∂ in 1786 but later abandoned it; Carl Gustav Jacob Jacobi reintroduced it in 1841. ESA ★ S001·14 · T001b 00:11:23
3. To find z_x, look only at the terms containing x; everything else is a constant, however many other variables there are. ESA ★★ T001b 00:15:08 · T001b 00:15:48
4. A function of x, y, t has three first partials z_x, z_y, z_t; for z_x treat y and t as constants. ESA ★ T001b 00:13:20
5. All rules of ordinary differentiation hold for partial derivatives: product, quotient, chain rule, implicit and composite functions. ESA ★★ T001b 00:13:57 · T001b 00:14:16

Why?

Why does every rule of ordinary differentiation still work?
With the other variables frozen, $f(x, y)$ is an ordinary function of x alone, and $\frac{\partial f}{\partial x}$ is its ordinary derivative (E4.1 is the usual limit with y fixed).

T001b 00:13:57 · S001·15

Exam traps

Exam trap. The partial symbol is ∂, not δ (the lecture stresses this) and not d. T001b 00:11:41

5 Second-order and mixed partial derivatives

Definitions

Mixed (crossed) partial derivatives. $z_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right)$ and $z_{yx} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right)$: differentiate with respect to one variable, then the other. S001·16 · T001b 00:18:24

Statements in the class material

> *"Note 1: The crossed or mixed partial derivatives are in general equal (Clairaut's theorem):* $\frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right)$*. That is, the order of differentiation is immaterial if the second order derivatives involved are continuous."* S001·17 · T001b 00:19:57

> *"Note 2: In the subscript notation, the subscripts are written in same order in which the differentiation is carried out, while in the* ∂ *notation the order is opposite. For example,* $\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = f_{xy}$*."* S001·17

Key points

1. Read z_{xyxx} left to right: $z \to z_x \to z_{xy} \to z_{xyx} \to z_{xyxx}$ (differentiate in x, then y, then x, then x). ESA ★ T001b 00:18:51 · T001b 00:19:21
2. Clairaut's theorem is "a very important theorem": whenever the second partials are continuous (true for all examination functions away from bad points), compute whichever mixed partial is easier. ESA ★★★ S001·17 · T001b 00:20:10
3. Verifying $f_{xy} = f_{yx}$ is a standard class-work question: x^y (CW P2) and $x^2 \tan^{-1}\frac{y}{x} - y^2 \tan^{-1}\frac{x}{y}$ (CW P8, W5.3). ESA ★★ CW U1 P2 · CW U1 P8

Exam traps

Exam trap. $\frac{\partial^2 z}{\partial y\, \partial x}$ means x first, then y: it is z_{xy}, not z_{yx}. S001·17

Exam trap. Clairaut needs continuous second partials: for $f = \frac{xy(x^2-y^2)}{x^2+y^2}$, $f(0,0) = 0$, one gets $f_{xy}(0,0) = -1$ but $f_{yx}(0,0) = 1$. (ADDED FOR CLARITY) S001·17

★ Past-paper questions

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. No question asks for a limit, continuity or the definition of a partial derivative. Computing partial derivatives is asked directly twice: Dec 2023 Q1.a (third-order partials of $e^x \log y + \cos y \log x$) and Jan 2024 set 4 Q1.c (find n for $\theta = t^n e^{-r^2/4t}$). Both are worked in the class 002 pack; this pack works the class-work problems of the same type (CW P4, P7, P8). Every other Unit 1 question (chain rule, Euler, Taylor, maxima and minima, Lagrange) uses the rules of this session.

No past-paper questions in scope.